# MESIO MVA exam 2025

Master in Statistics and Operations Research (MESIO UPC-UB)

Thursday 20th of March 2025, Room A5, 18:05-19:25

Only simple calculators are allowed.

# Instructions

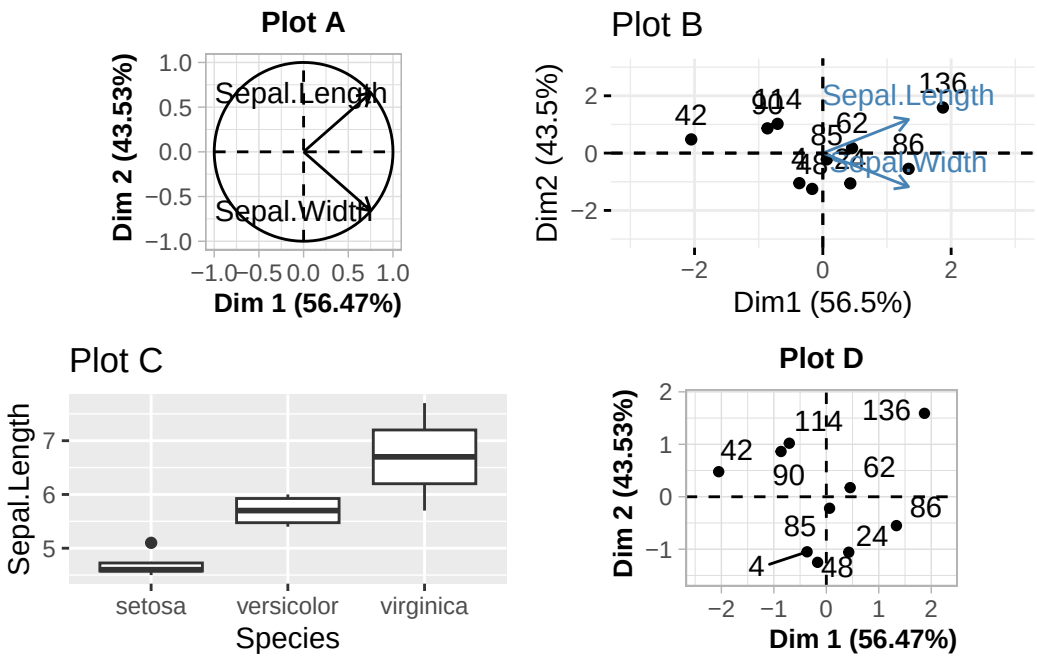
Answer the test questions in the attached sheet.

Answer the exercises on the official faculty exam sheet.

# Test

**(4.5 pts) Six multiple-choice questions, four options each: +0.75 for correct, -0.25 for incorrect.**

1. *Formula of the Mahalanobis distance of a point $x$ to the origin?*
   a. $D(x) = (\vec{x} - \vec{\mu})^t(\vec{x} - \vec{\mu})$
   b. $D(x) = (\vec{x} - \vec{\mu})^t\Sigma^{-1}(\vec{x} - \vec{\mu})$
   c. $D(x) = \sqrt{(\vec{x} - \vec{\mu})^t\Sigma^{-1}(\vec{x} - \vec{\mu})}$
   d. $D(x) = \sqrt{\vec{x}^t\vec{x}}$
2. *In which of the following spaces the Euclidean distances and the Mahalanobis distances coincide?*
   a. The space of the principal factor scores.
   b. The space of the scaled loadings.
   c. The space of $UD$ having $U$ the matrix of left singular vectors and $D$ the matrix of singular values.
   d. The space $F_s$, related to the standard axis in a PCA biplot.
3. *Which of the following representations is a biplot?*
   a. Plot C.
   b. Plot A.
   c. Plot D.
   d. Plot B.




4. We perform an SVD decomposition on a rectangular matrix $X_{n\times p}$ to get a rank 5 matrix that is an optimal approximation in the least square sense. Assume that $p > 5$ . Which of the following is correct?

   a. The goodness of fit is calculated as: $\frac{\sum_{i=6}^{p}\lambda_i}{\sum_{j=1}^{p}\lambda_j}$. With $\lambda_k$ being the $k$th eigenvalue from $X^tX$.

   b. It is not possible to get a rank 5 approximation.

   c. $\frac{\sum_{i=1}^{p}\sigma_i}{\sum_{j=1}^{p}\sigma_j}$ . With $\sigma_k$ being the $k$th singular value from $X$.

   d. None of the above.

5. *We have a distance matrix $D$ and perform metric MDS on it. Which of the following is correct?*

   a. It is not possible to have negative eigenvalues.

   b. We will obtain at most $p$ eigenvalues, with $p < (n-1)$ and $n$ being the total number of observations.

   c. We can get some negative eigenvalues but very close to 0. Those are computational rounding errors.

   d. Large negative eigenvalues indicates that the distance matrix was not calculated with the Euclidean distance.

6. *Given a contingency matrix $T$, one can always perform a ternary plot if:*

   a. Only if the number of rows, $I$, and the number of columns, $J$ , are equal.

   b. Only if $Rank(T) = 4 = I + J$, where $I$ is the number of rows and $J$ is the number of columns.

   c. Only if the number of columns or the number of rows in $T$ is 3.

   d. None of the above.

## Solutions

1. c
2. d
3. d
4. d
5. d
6. c

# Exercise 1

## Context

The data for this exercise has been published by the Food and Agriculture Organization of the United Nations (FAO). Some years ago, the FAO began reporting global, regional and country level indicators on the cost and affordability of a healthy diet (CoAHD). CoAHDs have been estimated on a yearly basis in 143 countries. They consists on measures of population's physical and economic access to least expensive locally available foods to meet the requirements for a healthy diet (Herforth et al. 2020, 2022).

The next table gives the cost of six food groups for 38 European countries in 2017.

## Data

The data is measured as purchasing power parity dollar per person per day (PPP) and contains the costs of individual food groups that make up the Healthy Die Basket:

- **CSTS**: Starchy staples (potatoes, corn and beans).
- **CASF**: Animal source foods.
- **CLNS**: Legumes, nuts and seeds.
- **CVEG**: Vegetables.
- **CFRT**: Fruits.
- **COFT**: Oils and fats.

There is also a categorical variable named **Group** with levels:

- *EU95*: Countries that joined the European Union until 1995.
- *EU17*: Countries that joined the EU between 1996 and 2017.
- *EEA*: Non-EU countries, members of the European Economic Area.
- *NEU*: "Genuine" non-EU countries.

*PPP: If a certain basket of goods costs $10 in the US and the equivalent basket costs 8000 units of a country's currency, the PPP exchange rate would be the number that converts 8000 units of that currency to $10, allowing for a comparison of the cost of living

```
[1] "Data:"

       Country Group  CSTS  CASF  CLNS  CVEG  CFRT  COFT
1      Albania   NEU 0.599 1.204 0.441 0.707 0.911 0.089
2      Austria  EU95 0.229 0.612 0.360 0.698 0.819 0.055
3      Belarus   NEU 0.506 0.826 0.643 0.349 0.678 0.174
4      Belgium  EU95 0.222 0.745 0.369 0.677 0.790 0.060
5     Bosnia/H   NEU 0.540 0.921 0.512 0.825 0.916 0.132
6     Bulgaria  EU17 0.504 0.963 0.478 0.774 0.959 0.101
7      Croatia  EU17 0.496 0.955 0.556 1.138 0.950 0.073
8      Czechia  EU17 0.297 0.700 0.377 0.757 0.693 0.076
9      Denmark  EU95 0.235 0.498 0.263 0.780 0.546 0.053
10     Estonia  EU17 0.333 0.752 0.671 0.542 0.742 0.085
11     Finland  EU95 0.195 0.543 0.407 0.727 0.597 0.077
12      France  EU95 0.246 0.640 0.277 0.848 0.859 0.066
13     Germany  EU95 0.235 0.666 0.301 0.816 0.714 0.052
```

```
14      Greece  EU95 0.410 0.797 0.410 0.742 0.581 0.096
15     Hungary  EU17 0.407 0.805 0.420 0.754 0.828 0.089
16     Iceland   EEA 0.277 0.596 0.277 0.525 0.484 0.055
17     Ireland  EU95 0.326 0.523 0.285 0.582 0.621 0.060
18       Italy  EU95 0.221 0.778 0.266 0.751 0.812 0.057
19      Latvia  EU17 0.335 0.794 0.314 0.921 0.648 0.111
20   Lithuania  EU17 0.368 0.702 0.343 0.834 0.670 0.086
21  Luxembourg  EU95 0.208 0.656 0.280 0.530 0.762 0.056
22       Malta  EU17 0.508 0.795 0.450 0.730 0.931 0.080
23  Montenegro   NEU 0.425 0.862 0.367 0.829 0.833 0.081
24 Netherlands  EU95 0.322 0.614 0.255 0.892 0.607 0.053
25  NMacedonia   NEU 0.525 0.949 0.347 0.712 0.640 0.146
26      Norway   EEA 0.565 0.691 0.257 1.122 0.615 0.075
27      Poland  EU17 0.318 0.584 0.377 0.890 0.651 0.090
28    Portugal  EU95 0.271 0.583 0.275 0.620 0.687 0.078
29    RMoldova   NEU 0.462 0.746 0.360 0.339 0.432 0.121
30     Romania  EU17 0.358 0.945 0.351 0.497 0.697 0.073
31      Russia   NEU 0.360 0.980 0.191 0.739 0.745 0.135
32      Serbia   NEU 0.572 0.998 0.612 0.920 0.872 0.096
33    Slovakia  EU17 0.350 0.667 0.311 0.974 0.634 0.077
34    Slovenia  EU17 0.290 0.698 0.346 0.722 0.669 0.072
35       Spain  EU95 0.278 0.623 0.211 0.739 0.796 0.053
36      Sweden  EU95 0.470 0.644 0.347 0.766 0.773 0.085
37 Switzerland   NEU 0.212 0.675 0.249 0.743 0.577 0.067
38          UK  EU95 0.147 0.494 0.195 0.399 0.533 0.053
```

```
[1] "Variables means:"
```

```
 CSTS  CASF  CLNS  CVEG  CFRT  COFT
0.358 0.743 0.362 0.734 0.718 0.083
```

```
[1] "Covariance matrix:"
```

```
      CSTS  CASF  CLNS   CVEG  CFRT   COFT
CSTS 0.015 0.014 0.008  0.005 0.006  0.002
CASF 0.014 0.026 0.009  0.004 0.012  0.003
CLNS 0.008 0.009 0.014  0.000 0.007  0.002
CVEG 0.005 0.004 0.000  0.032 0.008 -0.001
CFRT 0.006 0.012 0.007  0.008 0.018  0.000
COFT 0.002 0.003 0.002 -0.001 0.000  0.001
```

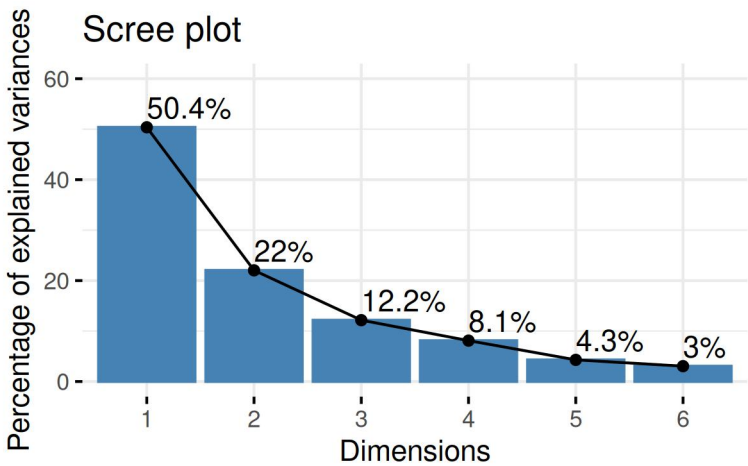
```
[1] "Correlation matrix:"
```

```
      CSTS  CASF   CLNS   CVEG  CFRT   COFT
CSTS 1.000 0.723  0.559  0.246 0.377  0.626
CASF 0.723 1.000  0.498  0.141 0.568  0.552
CLNS 0.559 0.498  1.000 -0.015 0.434  0.468
CVEG 0.246 0.141 -0.015  1.000 0.346 -0.144
CFRT 0.377 0.568  0.434  0.346 1.000  0.057
COFT 0.626 0.552  0.468 -0.144 0.057  1.000
```

## Questions (3.5 pts)

A PCA has been made on the correlation matrix using **Group** as supplementary categorical variable.

1. **Justify if there could be any notable difference in the PCA made on the covariance matrix. (0.35p)**
   There is a size effect on one of the variables (COFT has small variance compared to the others) as it has very low variance. Normalization of the data and use of PCA on the correlation matrix alleviates this problem. So there is a notable difference.
2. **Given the following scree plot, what should be the number of dimensions chosen for explain up to 80% of the variance? (0.35p)**




3 dimensions.

3. **Given the correlations between the standardized variables and the principal component scores, give interpretations for the three first principal components. Justify your answer. (0.35 pts)**

```
          Dim.1       Dim.2        Dim.3       Dim.4       Dim.5
CSTS 0.8782539 -0.04685628  0.294372159  0.03850180 -0.29246546
CASF 0.8783297  0.01988011 -0.002010915 -0.37745074 -0.10544874
CLNS 0.7511301 -0.15930404 -0.371311890  0.50918059 -0.02417026
CVEG 0.2210190  0.82267251  0.454901721  0.21855438  0.10575554
CFRT 0.6285236  0.55073867 -0.454720081 -0.18610073  0.16765814
COFT 0.6867885 -0.56004140  0.301980408 -0.02342064  0.34782416
```

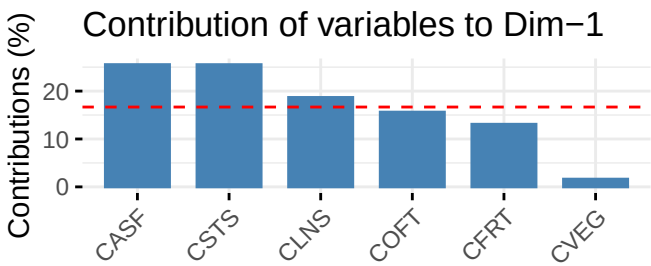
First dimension explained mainly by CSTS, CASF and CLNS. CFRT and COFT.

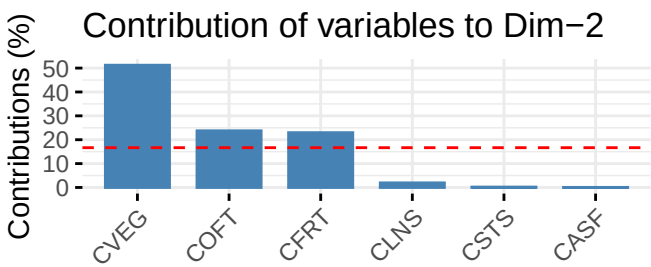
Second dimension explains CVEG. Also CFRT and COFT are explained. It is a contrast between vegetables and fruits wrt oils and fats.

Third dimension explains little about the original variables.

We can interpret the first principal component as those countries in which the cost of the basket does not comes from fruits and vegetables. While the second principal component is related to those countries in which the cost of the basket comes mainly by fruits and vegetables. The third PC cannot be interpreted clearly.

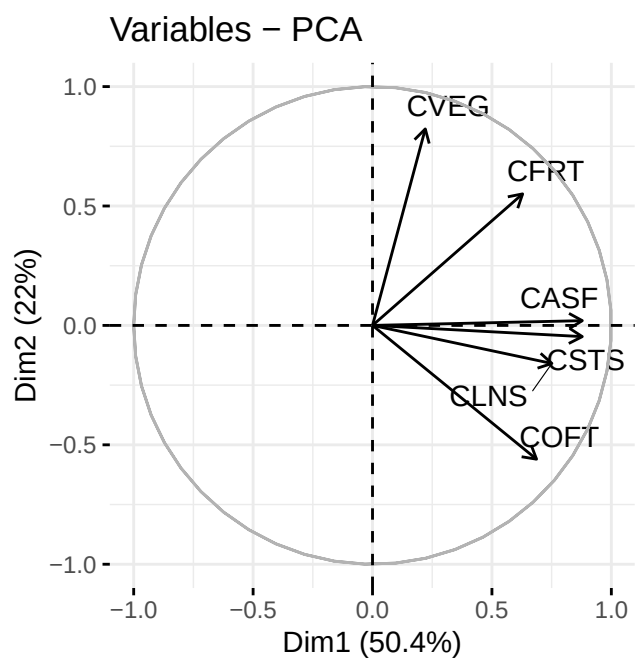
4. **Which information can be extracted from the following plots? Justify your answer. (The dotted line indicates the average value) (0.35 pts)**




These plots confirm the previous interpretations about PC1 and PC2 as we have the contribution of each variable variance captured by the principal components.

5. **Interpret the following plot. (0.35 pts)**




This plot shows the quality of the representation and the relations between the variables in terms of the factorial plane.
In general all vectors seems to have the same length with the exception of CLNS that is a little shorter than the others and so its representation is not as good as the other variables. CASF and CSTS are shown to be very high correlated and also correlated with CLNS.

CVEG and CFRT are also correlated but they appear to be not correlated with COFT.

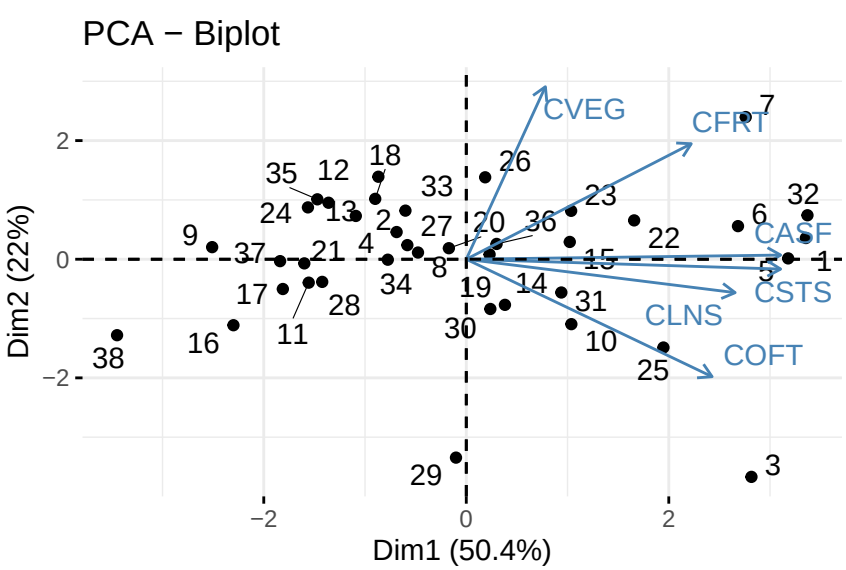
6. **Given the following biplot, interpret: (1 pts)**

   - **Individual 7 with respect to the factorial plane, the CFRT variable and the CSTS variable.**
     By the position in the factorial plane, individual 7 is probably an outlier that contributed highly to both principal components, meaning that is a country in which the basket has all type of food groups.
     Individual 7 is the one that is most explained with CFRT and the one that contributed the most as they align in the plot. For the CSTS the angle between the variable and the point indicates an ok representation.

   - **Individual 37 with respect to the factorial plane, the CVEG variable and the COFT variable.**
     Individual 37 lies on the first PC with negative values. This means that it does not contributes to the second PC and is interpreted mainly by the variables of the first PC in a negative way (below average value on these variables).
     The variable is not well represented wrt CVEG as the angle is nearly orthogonal. For the COFT variable we can say that the individual contributed in a negative way.

   - **What the individuals 13, 24 and 35 have in common?**
     They are all in similar positions so their interpretation wrt PC1 and PC2 are similar. In the context of the dataset these countries reflect a healthier basket.

   - **Why are they different to individuals 1, 5 and 32?**
     These countries are similar as they are positioned in similar positions on the factor plane. The difference is that this group represents a not so healthy basket.




7. **A factor analysis was performed using the maximum likelihood method to estimate the parameters for a two-factor model.(0.75 pts).**

   - **Identify the meaning of factor 1 and factor 2.**
     Factor 1 is more related to CSTS, CASF and COFT. More related to basket healthy products.
     Factor 2 is related to "CFRT", CVEG and "CASF". So it mainly represents fruits and vegetables

in the basket.

- **Give the factor analysis decomposition of the variance in matrix form (no calculations required).**
  Lt(L)+diag(specificities)
- **Are there variables not well approximated by this two factor model?**
  CVEG is not well represented. We can argue that also CLNS is not well represented as both present the largest specificities.

```
[1] "Uniquenesses (specificities):"

     CSTS      CASF      CLNS      CVEG      CFRT      COFT
0.3081683 0.2489401 0.5816483 0.7963394 0.2972415 0.1654631

[1] "Loadings:"


Loadings:
     Factor1 Factor2
CSTS  0.799   0.230
CASF  0.764   0.408
CLNS  0.602   0.238
CVEG          0.451
CFRT  0.309   0.779
COFT  0.870  -0.278

               Factor1 Factor2
SS loadings      2.438   1.164
Proportion Var   0.406   0.194
Cumulative Var   0.406   0.600
```

# Exercise 2

## Context

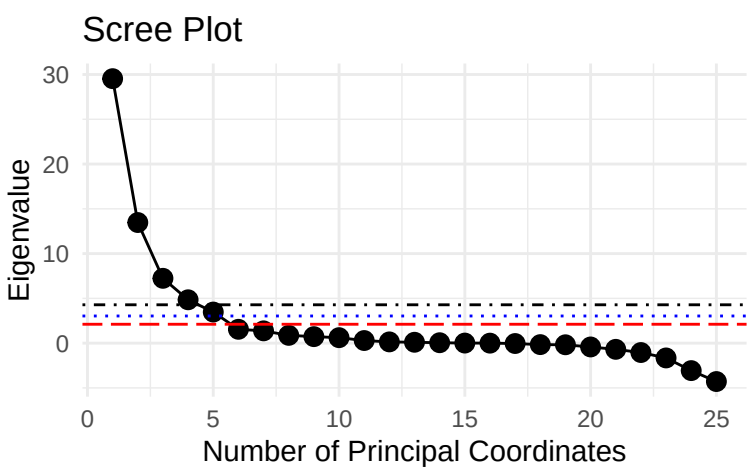
In this exercise we will explore data related to 25 U.S. universities. It consists on:

- **sat**: Average SAT score of entering freshmen.
- **top_fesh**: Percent of freshmen in top 10% of high school class.
- **app_acc**: Percent of applicants accepted.
- **sf_ratio**: Student-faculty ratio.
- **ann_expense**: Estimated annual expense.
- **grad_rate**: Graduation rate (%).

From the data, a distance matrix is given.

## Questions (2pts)

1. We apply metric MDS (Principal Coordinate Analysis) and obtain the following scree plot. Horizontal lines meaning:
   *Dot-dash line*: Average eigenvalue of the positive eigenvalues.
   *Dotted line*: Average eigenvalue of the absolute value of the eigenvalues.
   *Long dash line*: Average value of the eigenvalues.
   - Justify which could be a good number of dimensions with the available information. (0.75 pts).

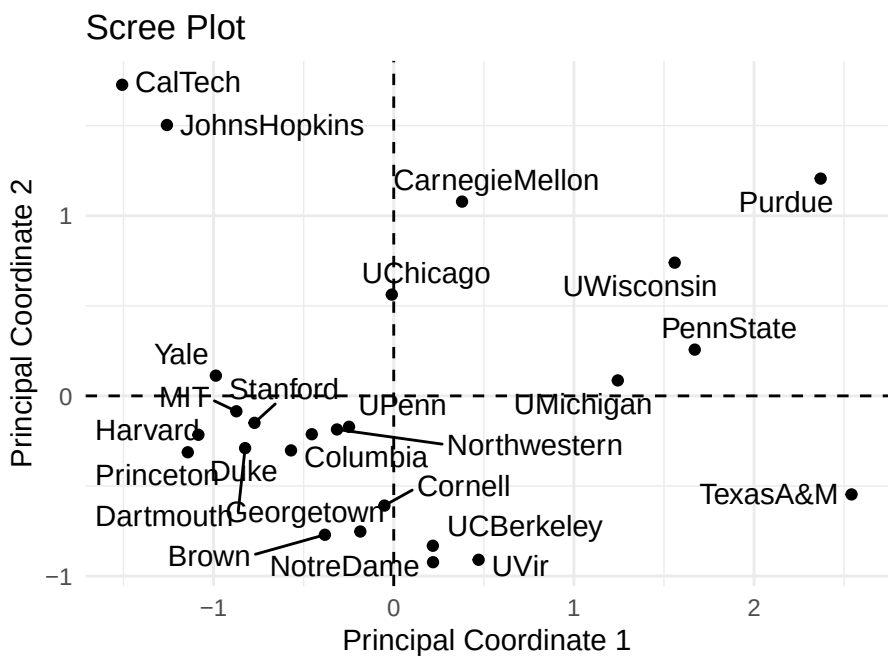



```
[1] "Eigenvalues information in descending order:"

      eigenvalue cumulative percentage
 [1,]     29.537     29.537     56.036
 [2,]     13.471     43.008     81.592
 [3,]      7.239     50.247     95.325
 [4,]      4.841     55.088    104.509
 [5,]      3.472     58.560    111.096
 [6,]      1.546     60.106    114.029
 [7,]      1.372     61.478    116.632
 [8,]      0.871     62.349    118.285
```

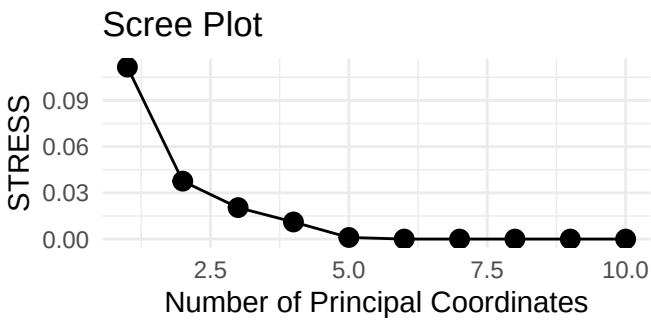
```
 [9,]      0.727     63.076    119.664
[10,]      0.612     63.688    120.825
[11,]      0.292     63.980    121.379
[12,]      0.139     64.119    121.643
[13,]      0.097     64.216    121.827
[14,]      0.047     64.263    121.916
[15,]      0.010     64.273    121.935
[16,]      0.000     64.273    121.935
[17,]     -0.043     64.230    121.853
[18,]     -0.164     64.066    121.542
[19,]     -0.182     63.884    121.197
[20,]     -0.427     63.457    120.387
[21,]     -0.692     62.765    119.074
[22,]     -1.047     61.718    117.088
[23,]     -1.654     60.064    113.950
[24,]     -3.066     56.998    108.133
[25,]     -4.287     52.711    100.000
```

Five dimensions seems a reasonable trade-off between the number of dimensions and the information explained, also it coincides with the average eigenvalue of the absolute value, one of the metrics explained in class for GoF criterion.



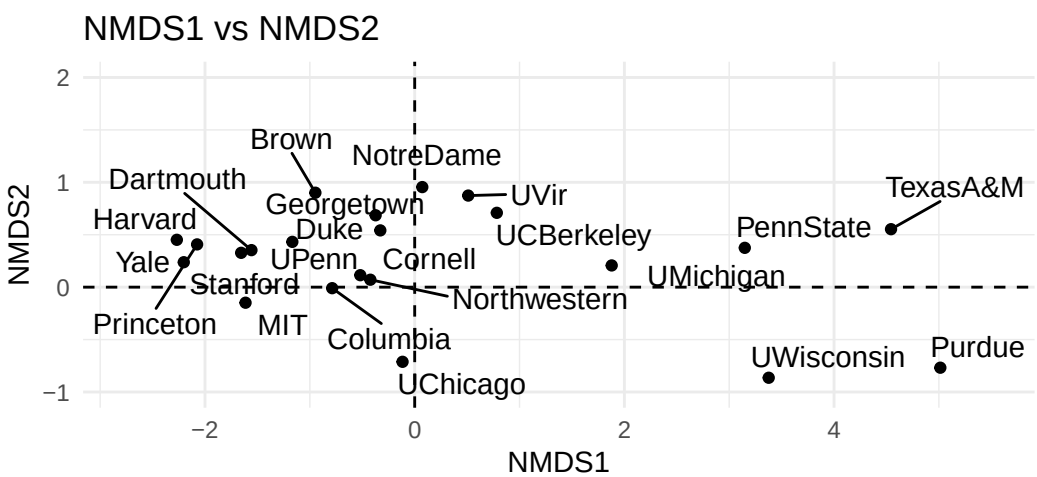

2. In addition, a non-metric MDS has been made. With the given information answer the following questions:

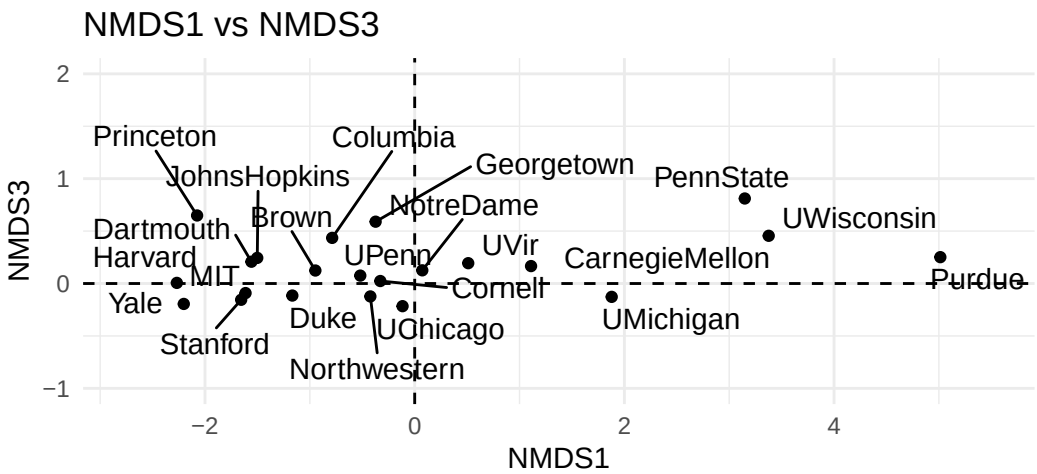
   - Choose the minimum number of Principal Coordinates such that the given approximation could be understood as a *Very good* approximation. Justify your answer. (0.5 pts).

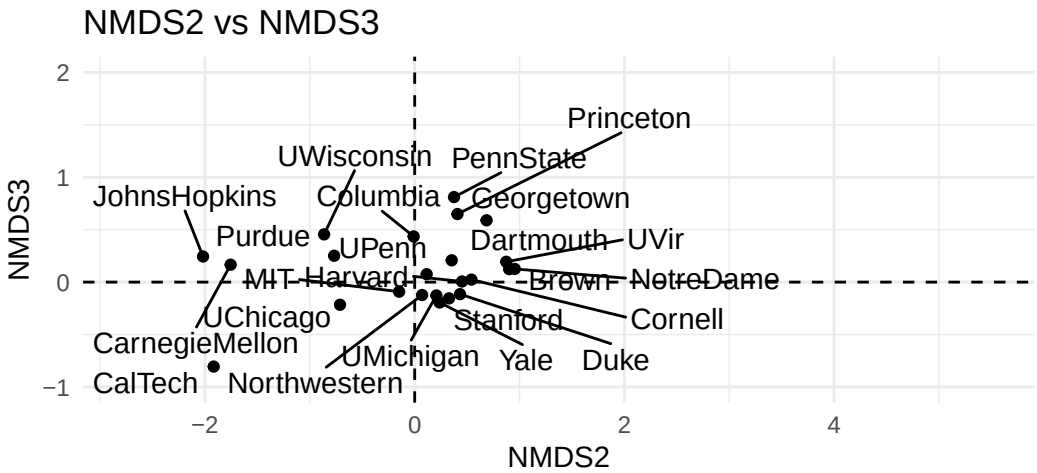



We can take 2 or 3 principal coordinates as those give less than 5% stress.

3. We decided to take 3 as the number of dimensions. Interpret the three Principal Coordinate axis with the aid of the correlations and the individuals plots.(0.75 pts)






| | MDS1 | MDS2 | MDS3 |
|---|---|---|---|
| sat | -0.981 | -0.017 | -0.087 |
| top_fresh | -0.914 | 0.204 | -0.314 |
| app_acc | 0.906 | -0.309 | -0.044 |
| sf_ratio | 0.846 | 0.395 | -0.258 |
| ann_expense | -0.789 | -0.527 | -0.119 |
| grad_rate | -0.807 | 0.456 | 0.261 |

PCoor1 and PCoor2 are the ones given better and easier interpretability.
In general we have public universities in the right and private ones in the left of the plot.
PCoor1 clearly makes a distinction between quality of the universities and the level of the students: Very high scores on SAT exam, high scores on first year, low probability of being accepted and high annual expenses.

PCoor2 is more related to a high annual expense in hand with a high graduate rate. But is not very significant.

PCoor3 has some correlation with top 10% of high school class as freshmen. But is not very significant.
Conclusion: PCoor1 is the principal coordinate that makes the distinction between the universities based on their characteristics.